Relatório Bayes 2

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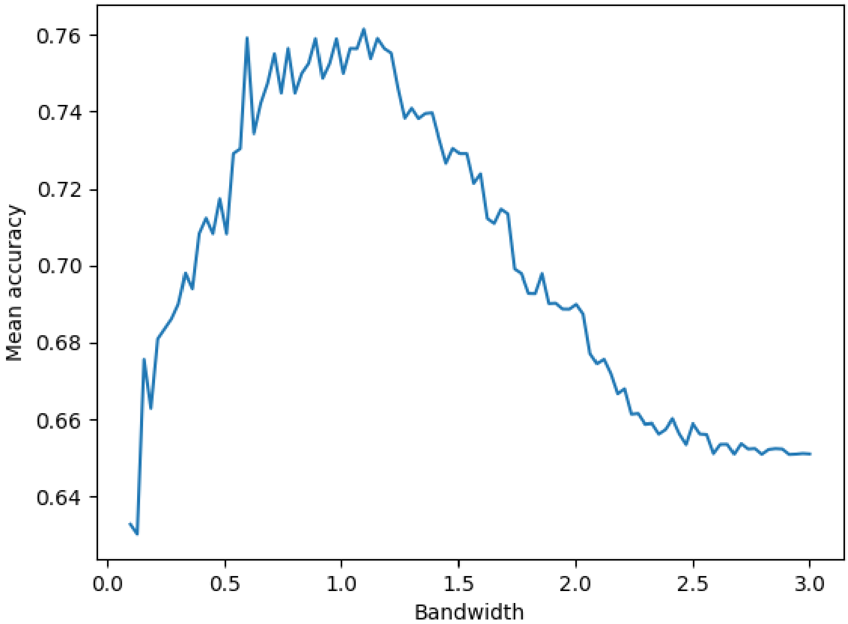

Figure 1: Accuracy vs Bandwidth

Max accuracy is achieved with a bandwidth of approximately 1.0.

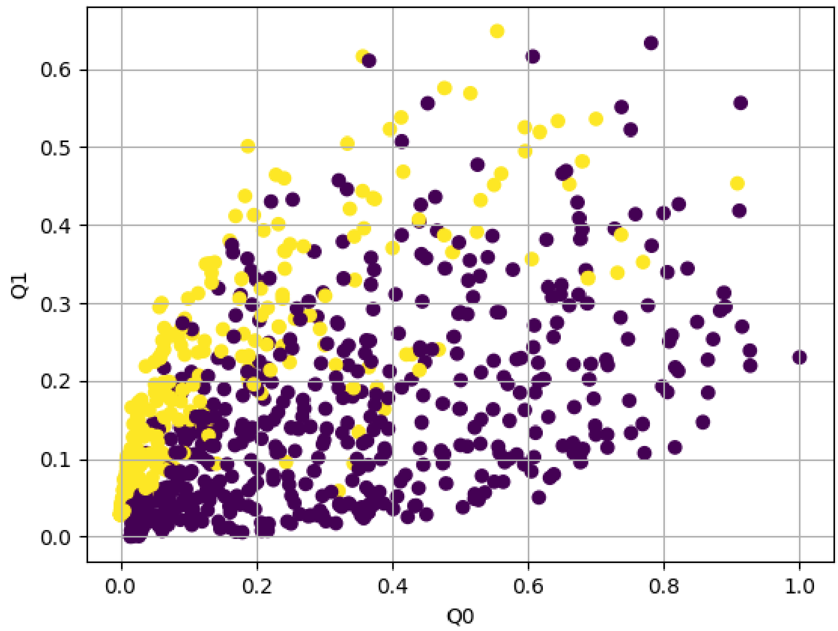

Figure 2: Best h=0.98

Visualy, the best accuracy is achieved when the samples are uniformly distributed in the likelihood space, with some overlap between the classes.

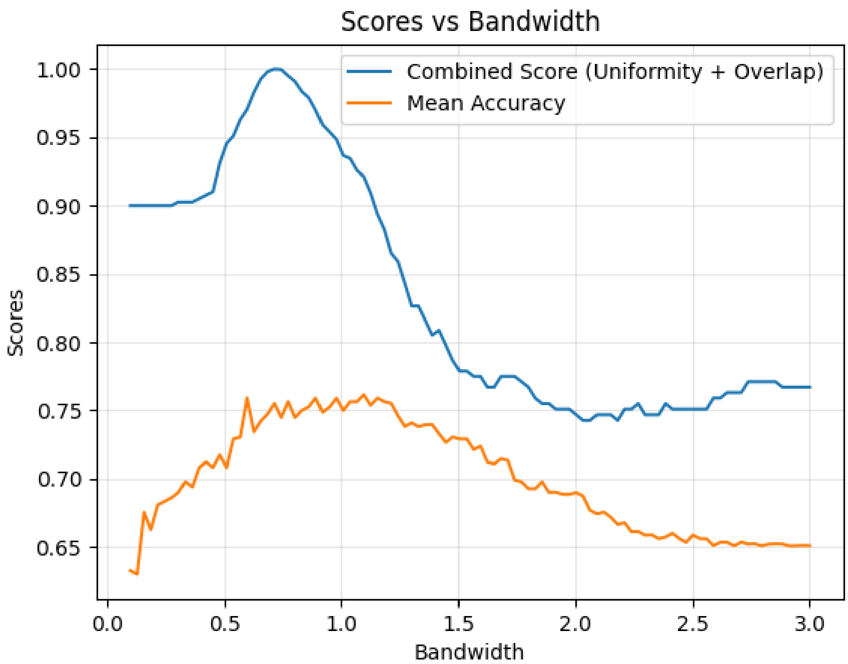

Figure 3: Combined Score vs Bandwidth

The combined score is defined as a weighted linear combination of two components: the uniformity score and the overlap score. Let $S_{\text{uniformity}}$ represent the uniformity score and S_{overlap} represent the overlap score. The combined score S_{combined} is given by:

$$S_{\text{combined}} = w_{\text{uniformity}} S_{\text{uniformity}} + w_{\text{overlap}} S_{\text{overlap}},$$

where $w_{\text{uniformity}}$ and w_{overlap} are the weights assigned to the uniformity and overlap scores, respectively, such that $w_{\text{uniformity}} + w_{\text{overlap}} = 1$.

The uniformity score $S_{\text{uniformity}}$ measures how evenly the samples are distributed within their respective regions in the likelihood space. It is computed as the proportion of non-empty bins in a discretized triangular region for each class, averaged over all classes:

$$S_{\text{uniformity}} = \frac{1}{2} \left(\frac{\text{Non-empty bins in Region 0}}{\text{Total bins in Region 0}} + \frac{\text{Non-empty bins in Region 1}}{\text{Total bins in Region 1}} \right).$$

The overlap score S_{overlap} quantifies the degree of overlap between the two classes in the likelihood space. It is defined as:

$$S_{\text{overlap}} = 1 - \frac{|O_{\text{actual}} - O_{\text{desired}}|}{O_{\text{desired}}},$$

where O_{actual} is the observed overlap proportion, O_{desired} is the desired overlap proportion, and the score is clipped to the range $[0, 1]$.

The combined score S_{combined} is designed to balance the trade-off between uniformity and overlap, with the weights $w_{\text{uniformity}}$ and w_{overlap} controlling the relative importance of these two factors. Empirically, it has been observed that S_{combined} correlates well with classification accuracy, making it a reliable predictor for model performance.